

AVERAGE

- ❑ The average of a set number is the value that represents the entire set.
- ❑ Averages are the measures of the central tendency of data.

$$\text{Average} = (\text{Sum of set of } N \text{ numbers})/N$$

- ❑ This also means, $\text{Average} \times N = \text{Sum of set of numbers}$

Let's suppose score of Sachin Tendulkar in last 5 matches are 90, 150, 10, 70, 45, then the average score will be:

A. 78

B. 365

C. 75

D. 73

$$\begin{aligned}\text{Average} &= (90 + 150 + 10 + 70 + 45)/5 \\ &= 365/5 \\ &= 73\end{aligned}$$

Ans D

1. The average age of students in section A of 40 students is 10 years and the average age of students in section B of 30 students is 12 years. Find the average age of students in both the sections taken together.
- A. $80/7$ B. $78/7$ C. $76/7$ D. $74/7$

2. The average of five consecutive numbers is 16. The highest of these numbers is:
- A. 21 B. 20 C. 18 D. 19
- 

3. The average age of a family of five people is 20 years. If the youngest member is 8 years old. What was the average age of the family at the birth time of the youngest member?

A. 12 Years

B. 15 Years

C. 18 Years

D. 16 Years

E. 23 Years

4. Average weight of Ram and Anil is 54 kgs. Average weight of Sunil, Vamsi and Krishna is 56 kgs. Find the average weight of Ram, Anil, Sunil, Vamsi, Krishna?

A. 56.3 kg

B. 55.2 kg

C. 55.8 kg

D. 56.4 kg



5. The average of 8 numbers is 12.5. Average of two of the numbers is 9.5. Average of another two numbers is 14.5. Find the average of remaining four numbers?
A. 13 B. 14.5 C. 13.5 D. 12
6. The average of 16 matches of a cricketer is 43.25. The average of 10 of the matches of the same cricketer is 41.5. Then find the average of the other six matches of the cricketer? (approximately)
A. 45.5 B. 46.17 C. 44.65 D. 46.25
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7. A man buys diesel of price ₹20, ₹22 and ₹24 per liter for three consecutive years. If he spends ₹5000 in each year, then find the average diesel price for him for all these three years? (approximately)
- A. 24.40 B. 22.65 C. 21.80 D. 22.40
8. The average visitors of a park on week days other than Sunday are 542. But on Sunday, the visitors are 948. In a month consists 31 days with 5 Sundays, find the average visitors of a day in the month? (approx.)
- A. 598 B. 604 C. 609 D. 607
- 

9. The average of 7 numbers is 32. If one number is included, average becomes 34. Find the number included?

A. 58

B. 42

C. 48

D. 54

10. The average of 9 numbers is 43. The average of first 5 numbers 42 and the average of last 5 numbers 46. Find the 5th number?

A. 53

B. 48

C. 52

D. 46



11. The average age of 20 boys is 13 years. The average age of first 11 boys is 12 years, average age of next 8 boys is 14 years. Find the age of 20th boy?
A. 12 B. 13 C. 14 D. 16
12. In an office 30 men and 20 women are working. The average salary of them is ₹3200. If women gets ₹300 less than men, find the salaries of a man, a woman.
A. ₹3400, ₹3100 B. ₹3500, ₹3200 C. ₹3320, ₹3020 D. ₹3350, ₹3050
- 

15. Average age of 4 members is 25 years. Age of first person is thrice the age of the second one. Age of the third person is $\frac{1}{4}$ of the age of the first one, age of the fourth person is $\frac{1}{3}$ of the age of 3rd one. Find the age of the 3rd one?
A. 15 years B. 12 years C. 16 years D. 18 years
16. Average income of Raju, Sonu and Rajani is ₹4600. Average income of Raju and Sonu is ₹4200. Average income of Sonu and Rajani ₹4800. Find the income of Raju?
A. ₹4600 B. ₹4320 C. ₹4200 D. ₹4400
- 

17. The average height of 25 boys in an academy of 40 is 165 cm. The average height of the remaining 15 is 155 cm. Find the average height of the academy?
A. 158 cm B. 160.50 cm C. 161.25 cm D. 162 cm
18. The average of 10.5, 16.5, 18, 13, 15, P is 14.5. Then find the value of P?
A. 14 B. 13.5 C. 15 D. 14.5
- 

19. Out of 5 numbers the average of the first 4 numbers is 26. And the average of the last four number is 25. If the first number is 20, then find the last number?
- A. 15 B. 16 C. 18 D. 20
20. The average weight of 16 students is 24 kg. If the weight of the class teacher is included, the average weight increases by 2 kg. Find the weight of teacher?
- A. 55 kg B. 52 kg C. 56 kg D. 58 kg
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21. The run rate of the first 15 overs of the cricket game is 4.8. If the target is 294. What should be the average run rate for the remaining 35 overs? (approx.)
A. 5.8 B. 6.3 C. 6.8 D. 6.2
22. The revenue of an American publisher for first 3 quarters is ₹13 Lakh, ₹12.6 Lakh and ₹14.3 Lakh. The average revenue for 4 quarters should be ₹15 Lakh. Find revenue target for the last quarter?(approx.)
A. 10 Lakh B. 15 lakh C. 20 lakh D. 25 lakh
- 

Equation based Question

31. The cost of 60 Apples and 100 guavas is Rs. 460. The cost of 100 apples and 60 guava is Rs. 500. Find the total cost of an apple and a guava.
- A. ₹8 B. ₹7 C. ₹5 D. ₹6
-
32. A bag contains a certain number of coins in the denomination of two-rupees and five-rupees. The total number of coins is 60, and the total value in the bag is ₹231. Find the number of five-rupee coins.
- A. 23 B. 36 C. 29 D. 37
- 

- 33. There are a certain number of hens and rabbits. The total number of heads is 50 and the total number of legs is 130. Find the number of hens.**
A. 15 B. 20 C. 25 D. 30
- 34. A box contains 90 nuts of 100 grams each and 100 bolts of 150 grams each. If the total weight of the box is 35.5 kg, find the weight of any empty box ?**
A. 10.0 kg B. 10.5 kg C. 11.0 kg D. 11.5 kg
- 

35. A railway half-ticket costs half the full fare, but the reservation charges are the same on a half-ticket as on a full ticket. One reserved first-class ticket from station A to station B costs Rs. 2125. Also, one reserved first-class ticket and one reserved half first class ticket from A to B cost Rs. 3200. Find the reservation charges for the ticket.
- A. ₹25 B. ₹30 C. ₹35 D. ₹45

36. A lady whose age is more than 50 years but less than 70 years, is celebrating her birthday, each of her sons has as many sons as brothers. The sum of the number of sons and grandsons is equal to the age of the lady. What is her age?
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37. Papa ate $\frac{1}{6}$ of a bunch of bananas. Then Mummy ate $\frac{1}{5}$ of the remaining bananas. Then brother ate $\frac{1}{4}$ of remaining and sister ate $\frac{1}{3}$ of remaining bananas. Everybody is eating banana, so I also ate $\frac{1}{2}$ of remaining bananas and left 6 bananas in a bunch. How many bananas were there in the bunch before papa ate ?

38. If $(m-4)$ is 7 more than n , then $(m+9)$ is how much more than $(n+2)$?
A. 12 B. 15 C. 18 D. 19

BLOOD RELATION

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BLOOD RELATION

- Introduction
- Generationwise Distribution
- Family Tree
- Types of Question

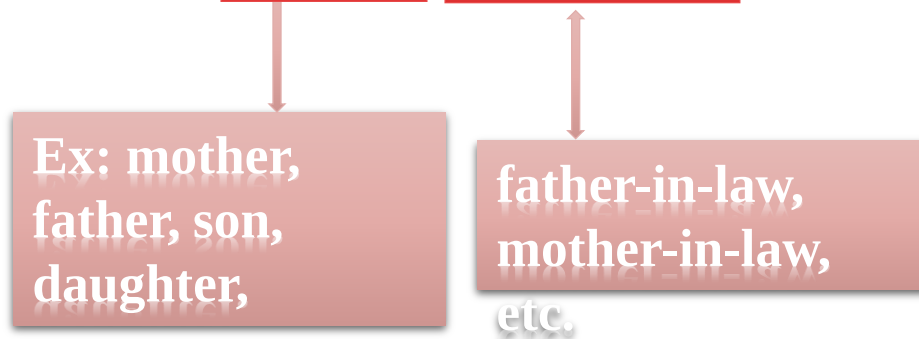
Type1:- Dialogue/ Conversation Based

Type 2:- Based on Puzzles

Type 3:- Coding-Decoding

BLOOD RELATION

Any relation in the world which either **by birth** or **by marriage** is called a Blood Relation.



This topic tests the analytical skills of the students.

How one can approach the solution of logical problems with the help of diagrams instead of calculations.

BLOOD RELATION

Relations of Paternal Side

Father's Father → Grandfather

Father's Mother → Grandmother

Father's Brother → Uncle

Father's Sister → Aunt

Children of Uncle → Cousin

Wife of Uncle → Aunt

Children of Aunt → Cousin

Husband of Aunt → Uncle

Relations of Maternal Side

Mother's Father → Maternal Grandfather

Mother's Mother → Maternal Grandmother

Mother's Brother → Maternal Uncle

Mother's Sister → Maternal Aunt

Wife of Maternal Uncle → Maternal Aunt

Children of Maternal Uncle → Cousin

FAMILY TREE

A family tree is a pictorial representation of genealogical data.

The following points help in drawing a family tree.

1. Represent female member by a **circle**.
 2. Represent male member by a **square**.
 3. The relation between two members of the family is shown by connecting a **double-headed arrow**.
 4. The spouse relation is represented by the two ends of a **two-sided arrow**.
 5. All the family members of the **upper generation** are represented **above** in the family tree.
 6. All the family members of the **same generation** are represented in the **middle** of the family tree.
 7. All the family members of the **next generation** are represented **below** in the family tree.
- 

GENERATIONWISE DISTRIBUTION OF RELATIONS

Generation I

Grandfather, Grandmother,
Maternal Grandfather, Maternal Grandmother



Generation II

Father, Mother, Uncle, Aunt
Maternal uncle, Maternal Aunt



Generation III

Self, Brother, Sister,
brother-in-law, sister-in-law



Generation IV

Son, daughter, nephew, niece

TYPES OF QUESTIONS ON BLOOD RELATION

1. Dialogue/ Conversation Based –

- ❖ In this type of questions, one person describes his/her relation with another person
- ❖ This may or may not be related to the person with whom the conversation is being made

2 Based on Puzzles –

- ❖ In this type, a puzzle is being given.
- ❖ A piece of brief information about multiple people being interrelated is given and sub-questions based on the same may be asked.

3. Coding-Decoding –

- ❖ The relationship between two people may be denoted using symbols.
- 

POINTS TO REMEMBER

- ❖ Do not assume the gender of the person based on the name/relation.

For eg. if in the question says X is the father of Y, the gender of Y cannot be determined unless mentioned in the question.

- ❖ Do In puzzle based questions, a web of relations can be formed, so do not solve such questions in a haste as the question seems lengthy.

- ❖ In case of coding-decoding blood relation, use a pictorial description to solve the question. This will make the symbols and relation more clear

Type1:- Dialogue/ Conversation Based –

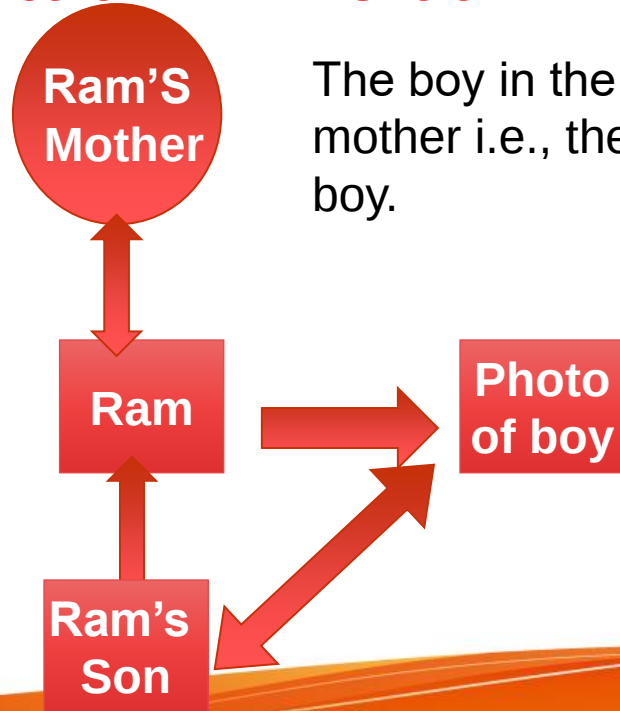
Pointing to a photograph of a boy Mr.Ram said, “He is the son of the only son of my mother.” How is Mr Ram related to that boy?

A. Brother

B. Uncle

C. Cousin

D. Father



The boy in the photograph is the only son of the son of Ram's mother i.e., the son of Ram. Hence, Ram is the father of the boy.

Hence, option D is the correct answer.

Type1:- Dialogue/ Conversation Based –

Pointing to a photograph, Rekha says to Lalli, "The girl in the photo is the second daughter of the wife of only son of the grandmother of my younger sister." How is this girl of photograph related to Rekha?

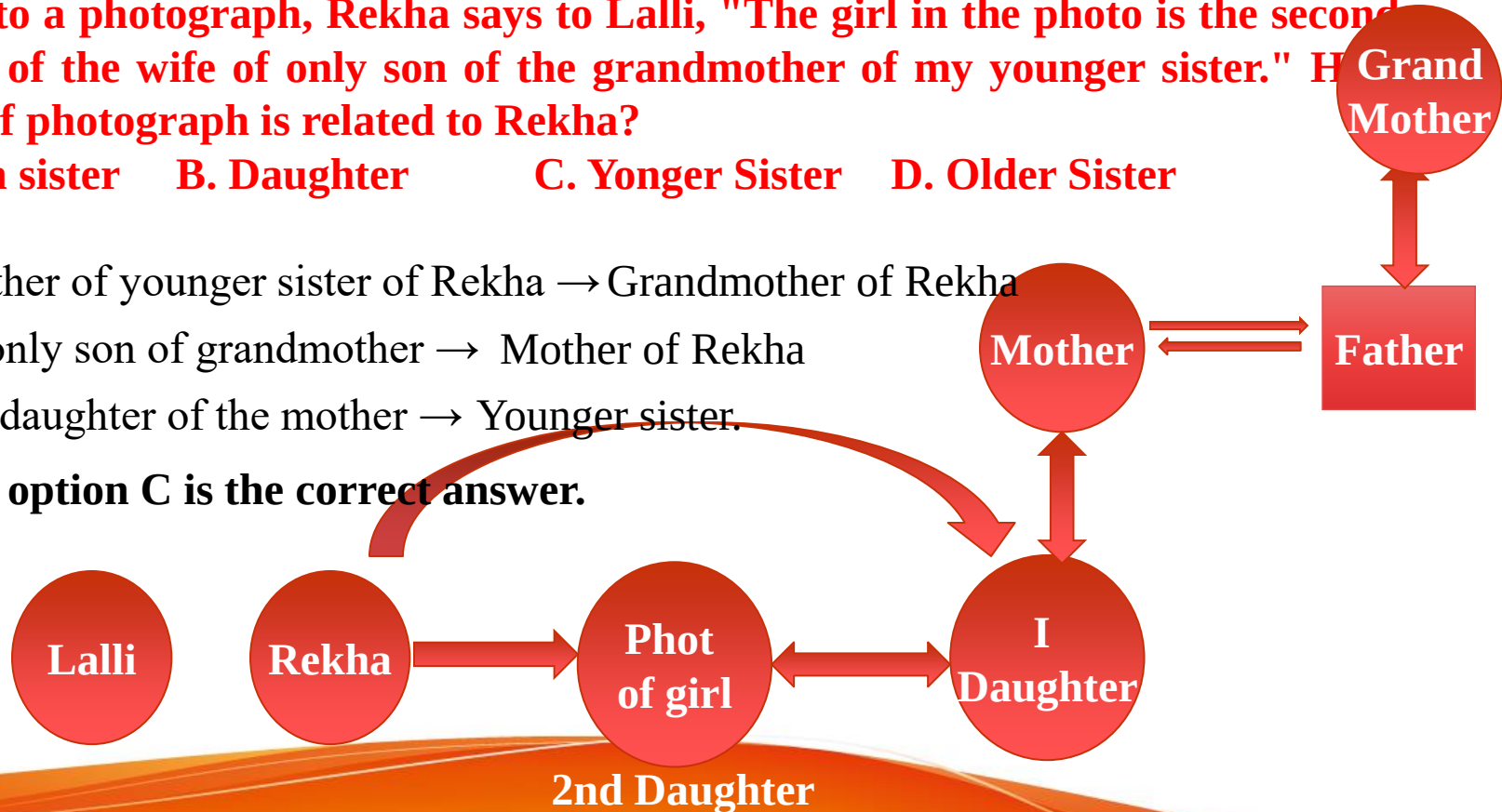
- A. Cousin sister B. Daughter C. Younger Sister D. Older Sister

Grandmother of younger sister of Rekha → Grandmother of Rekha

Wife of only son of grandmother → Mother of Rekha

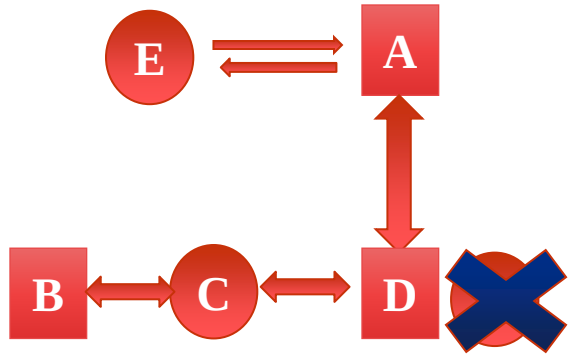
Younger daughter of the mother → Younger sister.

Hence, option C is the correct answer.



Type 2:- Based on Puzzles –

A has 3 children B is the brother of C and C is the sister of D E who is the wife of A is the mother of D There is only one daughter of the husband of E what is the relation between D and B?

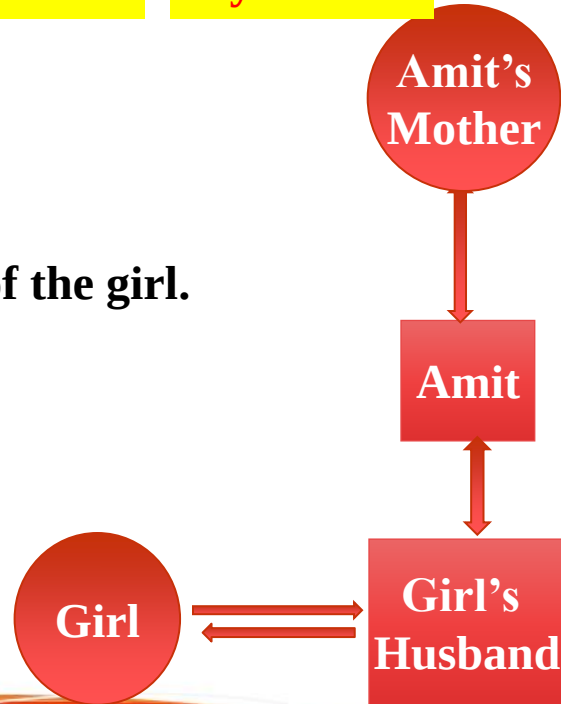


Therefore D is the brother of B.

Type 2:- Based on Puzzles –

Amit said - 'This girl is the wife of the grandson of my mother'. How is Amit related to the girl?

Therefore Amit is the Father-in-law of the girl.



Type 3:- Coding-Decoding

If $A + B$ means A is the mother of B ; $A \times B$ means A is the father of B ; $A \$ B$ means A is the brother of B and $A @ B$ means A is the sister of B then which of the following means P is the son of Q ?

A. $Q + R @ P @ N$

B. $Q + R \$ P @ N$

C. $Q \times R \$ P @ N$

$Q \times R \$ P \$ N$

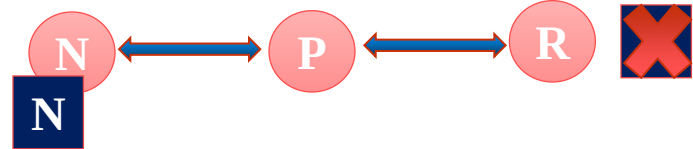
Option A

$Q + R @ P @ N$

$Q + R$ means
 Q is the mother of R

$R @ P$ means
 R is the Sister of P

Here P is daughter of Q
So, option A is incorrect.



Type 3:- Coding-Decoding

If $A + B$ means A is the mother of B ; $A \times B$ means A is the father of B ; $A \$ B$ means A is the brother of B and $A @ B$ means A is the sister of B then which of the following means P is the son of Q ?

A. $Q + R @ P @ N$

B. $Q + R \$ P @ N$

C. $Q \times R \$ P @ N$

D. $Q \times R \$ P \$ N$

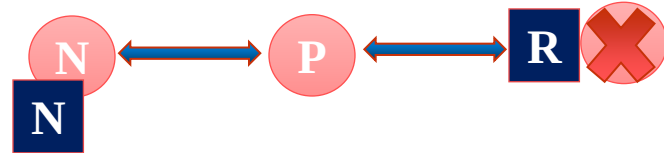
Option B

$Q + R \$ P @ N$

$Q + R$ means
 Q is the mother of R

$R \$ P$ means
 R is the brother of P

Here again, P is daughter of Q
So, option B is incorrect.



Type 3:- Coding-Decoding

If $A + B$ means A is the mother of B; $A \times B$ means A is the father of B; $A \$ B$ means A is the brother of B and $A @ B$ means A is the sister of B then which of the following means P is the son of Q?

A. $Q + R @ P @ N$

B. $Q \times R \$ P @ N$

C. $Q \times R \$ P @ N$

$Q \times R \$ P \$ N$

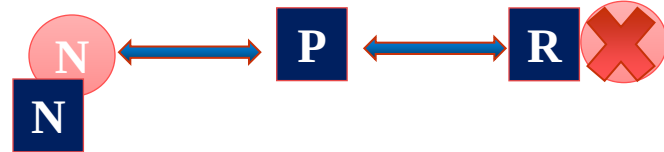
Option D

$Q \times R \$ P \$ N$

$Q \times R$ means
Q is the Father of R

$R \$ P$ means
R is the brother of P

Here again, P is son of Q
So, option D is correct.



Type 3:- Coding-Decoding

If $A + B$ means A is the sister of B; $A \times B$ means A is the wife of B, $A \% B$ means A is the father of B and $A - B$ means A is the brother of B.

Which of the following means T is the daughter of P?

A. $P \times Q \% R + S - T$

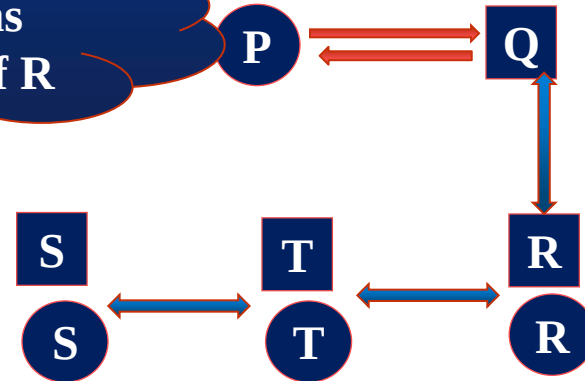
B. $P \times Q \% R - T + S$

C. $P \times Q \% R + T - S$

$P \times Q \% R + S + T$

Answer B

so $Q \% R$ means
Q is the father of R



5. Hitesh said to Nitin, “The girl playing with the football is the younger of the two sisters of the son of my mother’s husband.” How is the girl playing football related to Hitesh?
- A)Cousin B) Sister C) Daughter D) Sister-in-law E) Mother

6. Amit told Banita, “Yesterday I met the only brother of the daughter of my father’s mother.” Whom did Amit meet?
- A) Cousin B) Brother C) Nephew D) Father
E) Husband
- 

7. Pointing to a lady, Arun said “The son of her only brother is the brother of my wife”. How is the lady related to Arun?

A) Mother’s sister

B) Grandmother

C) Mother-in-law

D) Sister of father-in-law

E) Maternal aunt

8. Pointing to a person, Arun said to Sarika, “His mother the only daughter of your father.” How is Sarika related to that person?

A) Aunt

B) Daughter

C) Mother-in-law

D) Mother

E) Sister

- 9. If A's father is B, C is the father of B and D is A's mother, then how is C related to D?**
A) Father B) Grandfather C) Father – in – law D) Uncle E) Nephew
- 10. Mohan pointing to a girl in the photograph and said, “She is the daughter of the father of the sister of my brother.” How is that girl related to Mohan?**
A) Sister B) Wife C) Mother D) Aunt E) Daughter
- 

11. Pointing to a lady Gourav said, “the son of her only brother is the brother of my wife's sister”. How is the lady related to the Gourav?
- A) Mother B) Sister of Father-in-law
C) grandmother D) Aunt E) Can't be determined
12. Pointing to a person in the photograph, Arjun said, “His only brother is the father of daughter's father”. How is the person in the photograph related to Arjun?
- A) Father B) Uncle C) Cousin D) Brother-in-law
- 

All the 6 members of a family L, M, N, O, P, Q are traveling together. M is the son of N but N is not mother of M. L and N are married couple. P is brother of N. O is the daughter of L. Q is brother of M.

- 16. How many children does L have?**
A) One B) Two C) Three D) Four
- 17. How many male members are there in the family?**
A) One B) Two C) Three D) Four
- 18. Which of the following is a pair of females?**
A) LP B) MO C) OQ D) LO
- 19. Who is the wife of P?**
A) L B) Q C) M D) Cannot be determined
- 20. Who is the mother of M?**
A) O B) Q C) P D) L
- 